MARK SCHEME · TEACHER COPY

11BR · Topic 5: System Software

Cambridge AS Level Computer Science 9618

Paper 1 Theory Fundamentals

Total marks: 74

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Question 1 · May/June 2023 Paper 12, original Q5 (d) [7 marks]

Question	Answer	Marks
1(a)(i)	**1 mark** each to **max 5** • Manages the scheduling of processes // decides which order to run processes • Manages which resources the processes require • … such as allocating memory • Enables processes to share data • Prevents interference between processes // resolution of conflicts • Handles the process queue • It allows multi-tasking / multi-processing • ... by ensuring fair access, handling priorities **and** handling interrupts	**5**
1(a)(ii)	**1 mark** each to **max 2** • To help users to set-up / configure / analyse / optimise / maintain the computer ... • ... by for example, making memory allocation more efficient • ... by for example, checking the system for faults	**2**

Question 2 · May/June 2023 Paper 12, original Q7 (a),(b),(c) [13 marks]

Question	Answer	Marks

2(a)(i)	**1 mark** each to max 2 • Set of pre-written / pre-compiled / pre-tested subroutines • ... which can be **called** in other programs • ... by **installing/importing** the library	**2**
2(a)(ii)	**1 mark** for each bullet point. Mark in pairs; **1 mark** for a benefit and **1 mark** for an appropriate expansion • (main) memory requirements for program is reduced • ... as dynamic link library is loaded only once / when required • the executable file size of the program using the DLL will be smaller • ... because the executable does not contain (all) the library routines • maintenance not needed to be done by the programmer • ... because the DLL is separate from program • no need to **recompile** the main program when changes are made to DLL • ... because changes / improvements/ error correction to the DLL file code are done independently of the main program • A single DLL file can be made available to several application programs • ... Saving space in memory / easing the pressure on memory	**4**

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2(b)	No mark for choice. **1 mark** each to **max 3** for justification Interpreter • Allows the developer to make real-time changes • … so the program can be debugged **at each stage** • ... the effect of any changes made by the developer can be seen immediately • The developer can test when incomplete • … so small parts can be tested without having to test the rest of the program • … if one section does not work others can still be tested • To avoid dependent errors Compiler • The developer can debug multiple errors simultaneously • Produces an executable file • … so that the developer can test the program multiple times without recompiling	**3**

2(c)	**1 mark** each Breakpoints: • **Stop the code** at a specific line to check the current progress / values Dynamic syntax checks: • Highlight / underline / colour syntax errors **as the code is entered** Context-sensitive prompts: • Suggest the code to add // automatically complete statements Single stepping: • Run the code **one line at a time** so the values can be checked	**4**

Question 3 · May/June 2024 Paper 12, original Q6 [4 marks]

Question	Answer	Marks

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3	**1 mark** each to **max 4** **Max 3** marks for each management task: *Memory management:* **Max 3 marks** • Stores data from all currently running programs concurrently in RAM • Stops the data from overwriting each other in RAM/primary storage • Decides which processes should be in main memory • Makes efficient use of memory *Process management:* **Max 3 marks** • Allows one process to be paused whilst another process can be actioned • Decides which process is to be run next • Switches between processes to allow them to share the use of the processor • Identification/description of scheduling	**4**

Question 4 · May/June 2024 Paper 12, original Q8 (a),(b),(c) [10 marks]

Question	Answer	Marks
4(a)	**1 mark** each to **max 2**: • Creates an executable file • … so the code can be tested multiple times without having to recompile • … so **repeated testing** takes less time	**2**

4(b)

1 mark for identification of each feature and **1 mark** for matching description:

6

e.g.

For coding:

IDE feature	Description
Context-sensitive prompts	**Gives suggestions** for code as the user types instead of having to write/remember the code
Auto-correct	Corrects spelling mistakes so that user has fewer errors to correct

For presentation:

IDE feature	Description
Pretty-printing	Colour code keywords so the user can identify any errors
Expand/collapse (code) blocks	The user can hide code that they are not currently working on

For debugging:

IDE feature	Description
Single stepping	Run the code **one line at a time** // shows the effect of each line of code

	Breakpoints	Stop the code running at a set point to check the flow/variable contents		
4(c)	**1 mark** each to **max 2**: • Saves programming/testing time as code does not have to be written/re-written from scratch // code does not have to be tested • Code is already tested so it is more robust/likely to work • The programmer does not need to maintain the library // library routines are updated automatically • Can perform complex calculations that the programmer may be unable to do • Makes code more easily readable			**2**

Question 5 · May/June 2026 Paper 12, original Q7 (a),(b) [13 marks]

Question	Answer	Marks	Guidance
5(a)(i)	**1 mark** for each correct point to **max 2** • Efficient allocation of resources to processes • Moving processes between states • Allocating time slices to processes • Scheduling processes // An example of a scheduling algorithm • Resolution of conflicts when two processes require the same resource • Allows communication between processes	**2**	Do not accept allows multi-programming – that is a benefit not a task resource management is NE Accept checking priorities as scheduling Take care that answers do not refer to tasks done by the kernel or by memory management.
5(a)(ii)	**1 mark** for each correct point to **max 2** • Dynamically allocates RAM to processes • Frees up memory when the process closes • Makes sure two processes do not access the same memory • Accesses data from RAM when needed	**2**	Accept software or applications for processes in first mark point Take care that answers do not refer to tasks done by process management or by file management

5(a)(iii)	**1 mark** for each correct point to **max 3** • With continued use **files** can be split and stored in different locations on the disk so time is spent by the disk head moving between locations to access all parts of file • Defragmentation software moves split **files** into contiguous blocks // defragmentation software collects all the free space into a • contiguous block • … which means that the disk head does not need to move as much to access the next part of the file // … which means that the disk head does not need to move as much when writing a file • After defragmentation data **access/read/write** takes less time, so performance is improved	**3**	Note: The question asks how defragmentation improves computer performance. So just a description of the process is NE
5(b)(i)	**1 mark** for each correct point to **max 4,** mark in pairs • (Single) Stepping • …the student runs one line of code at a time **and** looks at the output/results/flow before moving to run the next line • Breakpoints • …These are inserted in the code to stop the code at a specific point **so that** the student can check the flow/values / test sub-sections of the code • (Variable) Watch window • …the student can view how/when the variables change during execution	**4**	Take care that the answer says how the student **uses** the tools and doesn't just describe each tool Award a maximum of two marks for names of debugging tools with no further suitable explanation Just check for errors is not enough for an explanation, that is what debugging means
5(b)(ii)	**1 mark** for each correct point to **max 2** • Partially compiled programs can be used on different platforms as they are interpreted when run • The code is optimised for the CPU as machine code is generated at run time • It is more efficient to run as the source code does not need recompiling	**2**	Note: Candidate responses need to answer the question why?

Question 6 · May/June 2026 Paper 13, original Q5 (a),(b),(c) [10 marks]

Question	Answer	Marks	Guidance
6	Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen.		

6(a)(i)	**1 mark** each to max 2 e.g. • Transfer data between peripheral devices • Translate data from I/O/peripheral devices • Loading and installation of device drivers • Hardware abstraction — hiding hardware differences from applications • Power management — sleep states, CPU throttling, battery control	**2**	
6(a)(ii)	**1 mark** each to max 2 e.g. • Memory management • Process management • File management • Security management // managing user accounts • Provide platform for running applications	**2**	
6(b)(i)	**1 mark** for: • Perform housekeeping/maintenance tasks // improving/optimising overall performance	**1**	
6(b)(ii)	**1 mark** each to **max 2** e.g. • Prepared a disk for future use by // initialising • …changing/setting file system // by example // directory structure // allocation tables • … naming the disk (volume label) • … creating a boot sector • …erasing all data • …creating partitions	**2**	
6(c)(i)	**1 mark** for: • Set of pre-written / pre-compiled / pre-tested routines	**1**	

6(c)(ii)	**1 mark** for benefit **1 for expansion** e.g. • Maintenance not needed to be done by the programmer • ... because the DLL is separate from program // maintained by library creator • The calling program does not need recompilation by the programmer when a DLL file changes • ... because the DLL file can be updated independently of the calling program • … updates will apply to all programs that use the DLL file • (Executable) file is smaller • ... because it does not contain (all) the library routines // code only linked at execution • (Main) memory requirements for program is reduced • … because the DLL files are only loaded into memory when required • ... one DLL file can be used by several running programs	**2**	

Question 7 · Oct/Nov 2023 Paper 13, original Q5 (a),(b) [9 marks]

Question	Answer	Marks
7(a)	**1 mark** for each bullet point (**max 2** for each feature). Context-sensitive prompts: • As the code is being written • …the options to complete the statement are shown Single stepping: • allows the programmer to execute the program one line at a time • …so that the effects of each statement can be seen	**4**

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7(b)(i)	**1 mark** for each bullet point (**max 2**). • To reduce the time it takes to download the program files from the web server // to upload them to the server in the first place • To reduce the amount of storage space used on the web server // user's device	**2**
7(b)(ii)	**1 mark** for each bullet point (**max 3**). • Disk needs to be prepared for initial use • Disk needs to be checked for errors • A new file system needs to be generated on the disk • The file allocation table needs to be set up	**3**

Question 8 · Oct/Nov 2025 Paper 13, original Q8 (a),(b),(d) [8 marks]

Question	Answer	Marks
8(a)	**1 mark** per bullet point, **max 2** • Large amounts of source code take time to compile • It can be slower to produce the object code than an interpreter • The code must be recompiled when it is changed • The program cannot run if there are errors • It is not possible to correct errors in real-time • One error can cause false reporting of multiple further errors • Sections of code / unfinished code cannot easily be tested	**2**

8(b)	**1 mark** per bullet point, **max 3** marks • Programming time is saved as code does not have to written from scratch • Testing time is saved as code is already tested / documented • A library routine is more likely to work, as code is already tested • Library routines automatically update if they are changed / improved • The programmer can use library routines to perform complex functions / procedures that they may not be able to write themselves	**3**
8(c)	**1 mark** for each bullet point, **max 3** marks • File management • Security management • Hardware / peripheral management // input / output management • Process management	**3**